

SREE LASYA BALAJI

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Available: January 2027 for Co-op or Internship

Education

MS in Computer Science | Northeastern University | Boston, MA

Courses: Database Management Systems, Machine Learning, AI Foundations

Jan 2026 – Expected May 2028

GPA: 3.6/4.0

B.Tech in Computer Science & Engineering | Sreenidhi Institute | Hyderabad, India

Courses: Data Structures & Algorithms, Computer Networking, Web Development, Operating Systems

Jun 2025

Technical Skills

Programming Languages: Python, SQL (Joins, Subqueries, Grouping), JavaScript, HTML, CSS

Backend & Frameworks: FastAPI, Django, Streamlit, REST APIs

Frontend: React, Next.js

Database Technologies: MySQL, MongoDB

ML Frameworks: Machine Learning (SVM, KNN, Naïve Bayes, Random Forest), TensorFlow, Computer Vision, OpenCV

Data Practices: Model Selection & Evaluation, NLP Preprocessing, Data Validation

Tools & Practices: Git/GitHub, Linux/Terminal, Modular Python Development, Agile/Scrum

Cloud & Certifications: AWS Academy Machine Learning Foundations, AWS Academy Data Engineering, AWS Academy Cloud Foundations, AWS Academy Cloud Architecting

Professional Experience

AI Instructional Assistant | CATLR – Northeastern University | Boston, MA

Sep 2026 – Present

- Partner with faculty leading **AI Curriculum Transformation** at the **D’Amore-McKim School of Business** and **College of Professional Studies** to build AI competencies and integrate **generative AI** into coursework.
- Follow a structured **prototype → test → evaluate → document → scale workflow** for every AI tool, using **Claude.ai** and other generative AI platforms to engineer reliable, reusable prompting strategies for real **course-assignment use cases**.
- Build a **repeatable, documented evaluation process** to validate AI tool outputs for **accuracy, safety, and reliability**, establishing clear criteria before any tool is recommended for real-world classroom use.
- Research how **peer institutions** and professions approach AI use in teaching, translating findings into **curriculum recommendations** for faculty stakeholders.
- Partner directly with **non-technical faculty stakeholders** to scope requirements and translate them into working, validated solutions, managing **multiple concurrent projects and deadlines**.

Projects

University Budget Management System (UBMS) — Full-Stack Application

- Designed a **23-table relational schema** in **MySQL 8** with **5 stored procedures**, **4 user-defined functions**, **5 triggers**, and **2 scheduled events** to automate financial workflows and enforce data integrity across a university’s complete budget operations.
- Built a **FastAPI** backend with **6 role-based API routers** and REST endpoints, supporting **5 distinct user roles** (Finance Officer, Admin, Dean, Registrar, Student), each with tailored data access and permissions.
- Developed role-specific dashboards in **Next.js 14/React/TypeScript** consuming live REST API data, giving each of the 5 roles real-time visibility into budgets, payroll, scholarships, and enrollment.
- Implemented **request validation and role-based access control** at the API layer to ensure each user role could only view or modify data relevant to their permissions.

Spam Mail Detection (Machine Learning)

- Preprocessed raw text data using **NLTK (tokenization, stop-word removal, stemming)** and vectorized features to prepare inputs for model training.
- Trained and compared **five classification models** (Naïve Bayes, SVM, KNN, Decision Tree, Random Forest), evaluating each on **accuracy and precision** to identify the most reliable approach.
- Built and deployed a **Streamlit web application** serving the trained model and vectorizer (via **pickle**) for real-time spam classification.

Road Safety Image Enhancement (OpenCV)

- Built a **Django web application** integrating real-time **object detection** with image enhancement techniques to improve visibility in rain, haze, and fog conditions.
- Applied **Dark Channel Prior** for haze removal, **guided filtering** for transmission refinement, and **CNN-based parameter tuning** for adaptive color and brightness restoration.
- Benchmarked multiple enhancement techniques against quantitative diagnostic metrics (**PSNR, SSIM**) to identify failure modes and determine the most reliable method for downstream object detection.